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ORIGINAL PAPER



Faulty use of the CIC-IDS 2017 dataset in information security research

Rohit Dube[1]

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Abstract
The summarized traffic flow version of the Canadian Institute for Cybersecurity Intrusion Detection Evaluation dataset created at the University of New Brunswick in 2017 is popular in the information security data science research community. Typically, researchers use the summarized data to develop supervised machine learning models and test the classification performance of these models. In this paper, we explore the adequacy of the summarized data for high-performance classification. We show that machine learning models developed over summarized data are unlikely to have practical import. Finally, we postulate that researchers may have a higher probability of creating a useful system if they use raw (non-summarized) data.

Keywords Machine learning · Classification · Network security · Intrusion detection system · Network traffic analysis

1 Introduction

The Canadian Insitute for Cybersecurity Intrusion Detection Evaluation dataset (CIC-IDS 2017) [1] has been available since 2017 and is heavily used by information security (infosec) data science researchers to develop and test machine learning models for intrusion detection. Typically, these infosec researchers use a high-level summarization of individual network traffic flows available as plain-text comma-separated values (CSV) files from the CIC-IDS dataset. Lower-level packet capture (PCAP) corresponding to the traffic flows is available but generally not used by infosec researchers because of the inconvenience of using large binary files containing byte-by-byte information.

This paper shows that summarized traffic flow data is inadequate for training machine learning models for intrusion detection in production networks. As a result, the value of infosec research that uses the summarized data is suspect.

The rest of this paper is organized as follows: Sect. 2 motivates the subject of this paper and discusses related work. Section 3 presents a summary of the CIC-IDS project, highlighting the aspects relevant to intrusion detection machine learning models. Section 4 discusses deployed intrusion detection/prevention systems (IDS/IPS) and sets up the "hurdle" that machine learning models need to clear. Section 5 extracts the most valuable features from the CIC-IDS CSV files. Section 6 outlines the limits to the signal contained in these "most valuable" features. Finally, Sect. 7 concludes the paper with some hints for future work.

2 Related work

The official reference for the CIC-IDS dataset—[1]—has 2815 Google Scholar references [2]. [1], and many of the research papers that cite it describe machine learning models tested over the CSV files. The citing papers themselves have a large number of references. For example, [3] describes AdaBoost trained over the CSV files and has 174 citations. [4] describes neural network models trained over the CSV files (and other data) and has 1160 citations.

None of the classification papers that use the CIC-IDS dataset address the value of classifying traffic flows using the summary information in CSV files. Further, these papers compare the performance of various classification techniques over the CSV files using measures such as accuracy, precision, recall, and F1 score. However, they do not compare the performance of their machine learning models with that of (usually non-machine-learning) opensource or commercial IDS/IPS [5–9].

[1] The CSV files take up 863MB of space and are significantly more convenient to use than PCAP data, which takes up 48GB of space.

✉ Rohit Dube
rdube@vmware.com

[1] Networking and Advanced Security Business Group, VMware Inc., 3401 Hillview Ave., Palo Alto, CA 94304, USA

Panigrahi and Borah [10] points out some problems with the CSV files. The paper's authors notice a high imbalance between the various traffic flow classes and that some columns (features) in the CSV files have missing values.

The research described in [11] and [12] appears to have been conducted in parallel with our work. The papers' authors point out several problems in the generation of the CSV files from the PCAP data. The authors find that the errors lead to mislabeling of a notable fraction of the traffic flows. This mislabeling calls into question the myriad machine learning approaches built on top of the CSV files.

Rosayand et al. [13] is another paper that describes research conducted in parallel to our own. Similar to [11] and [12], the paper's authors discover problems in the summarization of PCAP data into CSV files. Further, they describe a tool that fixes the summarization. Finally, they re-run the machine learning algorithms previously discussed in [1] and obtain better classification results over the fixed data.

Panigrahi and Borah [10], Engelen et al. [11], Liu et al. [12] and Rosayand et al. [13] all get at problems in the CSV files. However, the papers do not address the use of CSV files versus PCAP data to compare machine learning models.

We are unaware of any research comparing the performance of machine learning models developed over the CSV files with that of IDS/IPS deployed in production networks. This deficiency in existing literature is addressed (up to a point) in our work.

It is worth contrasting the use of the CIC-IDS dataset with KDD-CUP99 and its derivate NSL-KDD. Both are older datasets—from 1999 and 2009, respectively—that are still in use by infosec researchers. The official reference for KDD-CUP99—[14]—has 365 Google Scholar citations whereas the one for NSL-KDD—[15]—has 4830 citations [2]. Given the CIC-IDS dataset's relative age and citation count (between that of KDD-CUP99 and NSL-KDD), one can conclude that the CIC-IDS dataset is an important resource for infosec researchers, and scrutinizing this dataset constitutes a worthy research activity in itself.

3 CIC-IDS background

The CIC-IDS data is generated over a network of computers that emulate victim and attacker nodes. Multiple attack scenarios are enacted over the network for five consecutive days against a backdrop of normal network traffic.

3.1 Data generation

The CIC-IDS team generated "benign" network traffic by initiating normal traffic among the victims. Then, they initiated 14 well-known attack types between attackers and victims using publicly available tools. Finally, they saved the PCAP data corresponding to all traffic flows.

Subsequently, the CIC-IDS team extracted traffic flow "features" (summarized parameters of the individual traffic flows) from the PCAP data, saving the features into CSV files along with the traffic flows' label (benign or one of 14 attack traffic types). The publicly available CSV files have 78 traffic features (see table 1 for a sample) and 1 label for each traffic flow.

Table 1 Sample summary features of traffic flows

Feature	Description
Destination	Destination TCP or UDP port of the
Port	The first packet in the flow
Fwd Packet Length Std	Standard deviation of packet length in forward direction
Min Packet Length	Minimum length of a packet
Packet Length Variance	Variance of packet length
PSH Flag Count	Number of packets with the PSH flag set in the TCP header
Active Max	Maximum time a flow was active before becoming idle

3.2 Attack traffic flows

All traffic flow types—15 of them—are summarized in table 2. Note that there is a category for benign traffic flows and 14 categories for attack traffic flows.

All 14 of the attack types are "well-known" to the infosec community [1]. As such, organizations deploying IDS/IPS systems expect protection against these types of attacks. A part of the reason for this expectation is that legacy systems persist in many organizations, even after these organizations have lost the ability to "patch" known vulnerabilities on the systems. Another reason is that well-known attacks are better documented, baked into available hacking tools, and more accessible to new attackers than new or esoteric attacks.

There are two additional significant points to note about the attacks. First, all attacks are deterministic, and their characteristics are determined entirely by the attack tool and configuration. The tool used may allow some variation between multiple runs of an attack, but the traffic flow would be within the envelope circumscribed by the tool. Second, the CSV files record the summary of each attack traffic flow but do not retain the underlying sequence of bytes and packets.

Table 2 Labelled traffic flows in CIC-IDS CVS files

Num.	Traffic flow	Description
15	DDoS	DDoS attack using the tool LOIC
14	PortScan	Portscan using the tool LOIC
13	Bot	Botnet emulation using the tool Ares
12	Infiltration	Emulate infiltration using Metasploit
11	Web Attack Sql Injection	Exploit vulnerable application DVWA using SQL Injection
10	Web Attack XSS	Exploit vulnerable application DVWA using XSS
9	Web Attack Brute Force	Exploit vulnerable application DVWA using Brute Force
8	Heartbleed	Exploit Heartbleed bug using the tool Heartleech
7	DoS GoldenEye	DoS attack using the tool GoldenEye
6	DoS Hulk	DoS attack using the tool Hulk
5	DoS Slowhttptest	DoS attack using the tool Slowhttptest
4	DoS slowloris	DoS attack using the tool Slowloris
3	SSH-Patator	Brute force SSH password cracking using the tool Patator
2	FTP-Patator	Brute force SSH password cracking using the tool Patator
1	Benign	Normal traffic

3.3 Shortcomings of CSV Files

The CSV files do not contain IP addresses (or equivalent identifiers) for the source or the destination. Such addresses are useful in determining misbehaving and malicious nodes.

Also, the CSV files do not contain the IP protocol identifier or the source TCP/UDP port number. Such information can be useful in determining some types of attacks—especially when coupled with the source IP address.

Finally, the CSV files do not include any timestamps (except for coarse information such as the day of the traffic flow that can be extracted from the CSV filename). Timestamps, when coupled with information on the source of the traffic flow, can help determine the intensity of communication. In turn, the intensity of communication can help detect malicious nodes.

4 Infosec research versus traditional IDS/IPS

Since the 14 attacks listed above are deterministic, the attacks can be detected and prevented using traditional IDS/IPS techniques:

- Protocol decoding to detect malformed network protocol headers and flag rarely used network protocol options.
- Regular expression-based byte-stream signatures derived from previous attacks to classify new traffic flows.
- IP address reputation to detect known malicious network nodes.

Most traditional IDS/IPS systems could detect these (known) attacks with close to 100% accuracy and close to 0 false positives and false negatives.

For machine learning models to be useful, one or more of the following would have to be true:

1. Classification performance of the machine learning models is higher than that of traditional IDS/IPS.
2. The cycle time to train a new machine learning system to address a previously unknown attack and deploy it is faster than the cycle time to create and deploy a new IDS/IPS signature for the attack.
3. A deployed machine learning system is significantly easier to maintain than a deployed signature-based IDS/IPS.

Unfortunately, because the infosec researchers do not compare their machine learning models to traditional IDS/IPS, they don't address any of the methods of proving superiority over traditional IDS/IPS. Given traditional IDS/IPS's high classification performance for known attacks, it is hard to imagine any other technique being much better. As for a reduction in cycle time for creating a defense against a previously unknown attack—since the machine learning models discussed by the infosec researchers use supervised learning, there is no reason to believe that collecting and labeling attack traffic flows would be faster than examining the same traffic flows to create IDS/IPS signatures. Finally, since none of the machine learning models discussed above were deployed (even over a test network), there is no body of work indicating that such systems are easy to maintain.

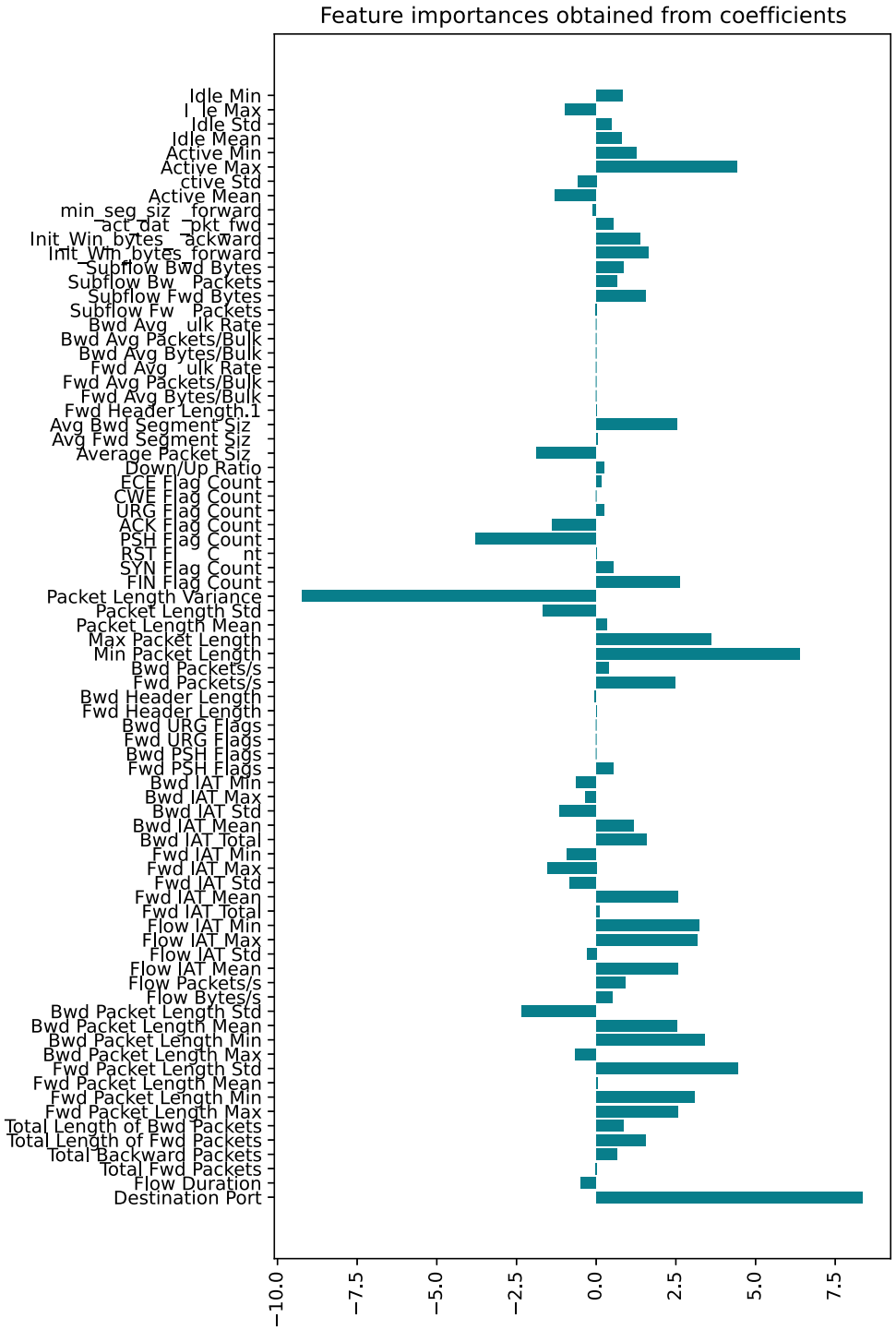


Fig. 1 Logistic regression coefficient-by-feature

5 Feature analysis

Notwithstanding the critique of infosec research in the previous section, we wanted to understand the signal in the summary features used by the machine learning models.

Toward this, we created a new CSV file that included the benign traffic flows from the first day of the CIC-IDS experiment and all of the attack traffic flows over the five days. Our new CSV file contains an equal proportion of benign and attack traffic flows: 529,918 flows for class 1 and 557,646 flows for classes 2–15.

Then, we run logistic regression for 100 iterations and record the coefficient for each of the 78 features (see figure 1). We extract the 6 features with the largest absolute coefficient value for further study. These 6 features are the same as in table 1.

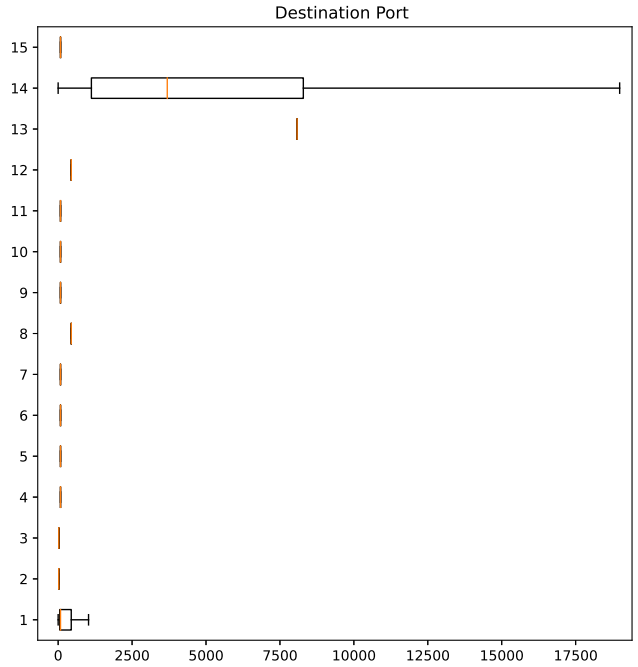


Fig. 2 "Destination Port" range by class

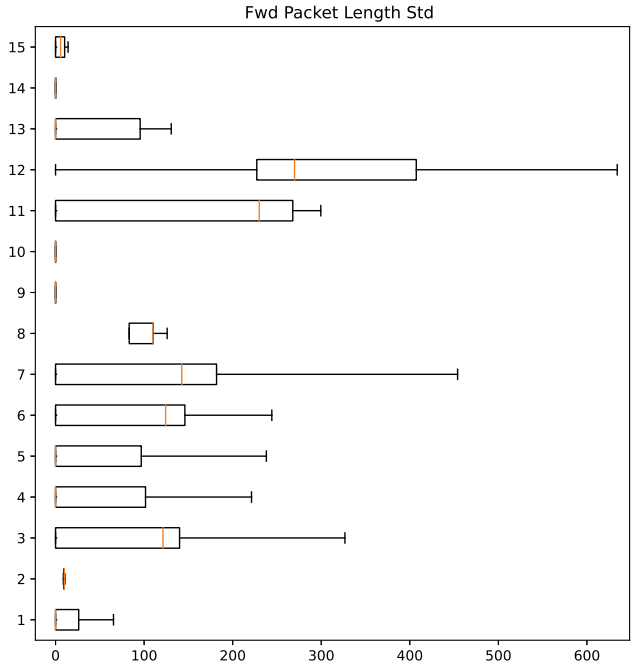


Fig. 3 "Fwd Packet Length Std" range by class

As a sanity check, we re-run logistic regression with just these 6 features for 100 iterations. We notice that the 78-feature logistic regression produces a weighted precision of 0.973 at a recall of 0.975. The 6-feature logistic regression produces a weighted precision of 0.872 at a recall of 0.888. We conclude that most of the signal in the CSV files is concentrated in these 6 features. (Note that the weighted precision and recall scores take into account the true number of occurrences of each class in the data.)

We also investigated principal component analysis (PCA) as an alternative to logistic regression to determine the importance of individual features. However, because PCA ranks components that are composites of the features in the CSV files, the results are difficult to interpret. We leave out the PCA results from this paper in the interest of brevity.

5.1 Destination port

Figure 2 is a "box and whiskers" plot that shows the range of values of the Destination Port across all classes.[1] We see that the range for class 1 (benign) includes the range for all other classes except classes 13 and 14. [2]

Since class 13 is for a bot, we understand that the attack tool chose a specific port for bot communication. Also, since class 14 is for a portscan, we can appreciate that the attack tool engages many destination ports. Finally, the median port number for class 1 seems indistinguishable from that of several attack classes.

From the above, we conclude that while the destination port is useful to signal the presence of classes 13 and 14, it is likely not useful for detecting the other attack classes.

5.2 Fwd Packet Length Std

Figure 3 shows the range of values for "Fwd Packet Length Std" (standard deviation of packet length in the forward direction) across classes. The range for class 1 subsumes the range for some attack classes (2, 9, 10, 14, 15) but is, in turn, subsumed by the range of several other attack classes. Further, the overlap in ranges between the attack classes indicates that distinguishing between classes using "Fwd Packet Length Std" alone would be difficult.

5.3 Min Packet Length

Figure 4 shows the range of values for "Min Packet Length" (minimum length of a packet) across classes. The range for class 1 subsumes the range for all the attack classes. Further, the range for several attack classes—2, 3, 4, 5, 6, 7, 8, 9, 10, 11—appears identical. It does not seem likely that the "Min Packet Length" feature would be useful on its own to distinguish between the traffic classes.

[1] The orange line in the "box and whiskers plot" represents the median, the box represents the 25-percentile to the 75-percentile range, and the whiskers represent the 0-percentile to the 100-percentile range. We abbreviate the name of the plot to just "box plot" in the rest of the paper.

[2] The first column in table 2 is the class number.

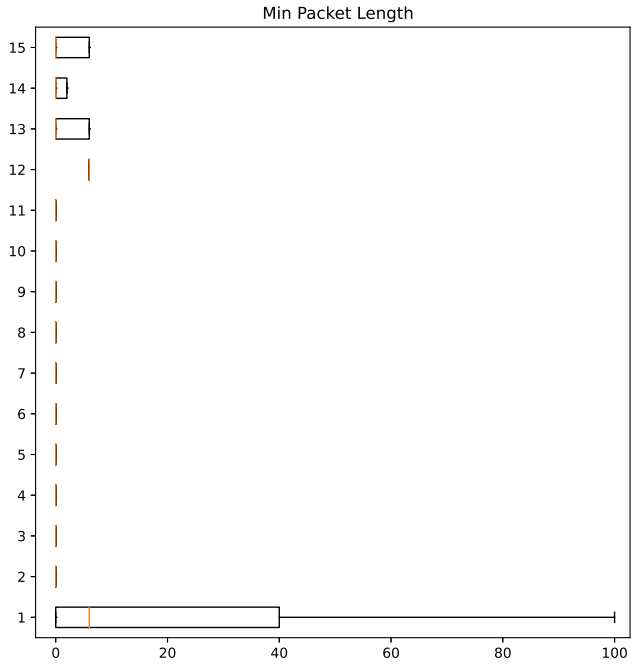


Fig. 4 "Min Packet Length" range by class

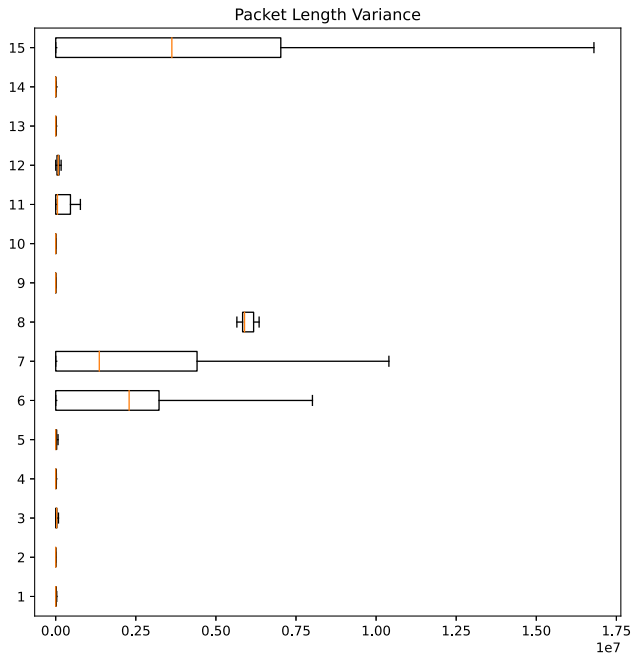


Fig. 5 "Packet Length Variance" range by class

5.4 Packet Length Variance

Figure 5 shows the range of values for "Packet Length Variance" (variance of packet length) across classes. Here, the ranges for a majority of the classes appear identical. Further, the ranges for classes 6, 7, 11, and 15 subsume the ranges of the other classes. Again, on its own, "Packet Length Variance" does not appear likely to distinguish between traffic classes.

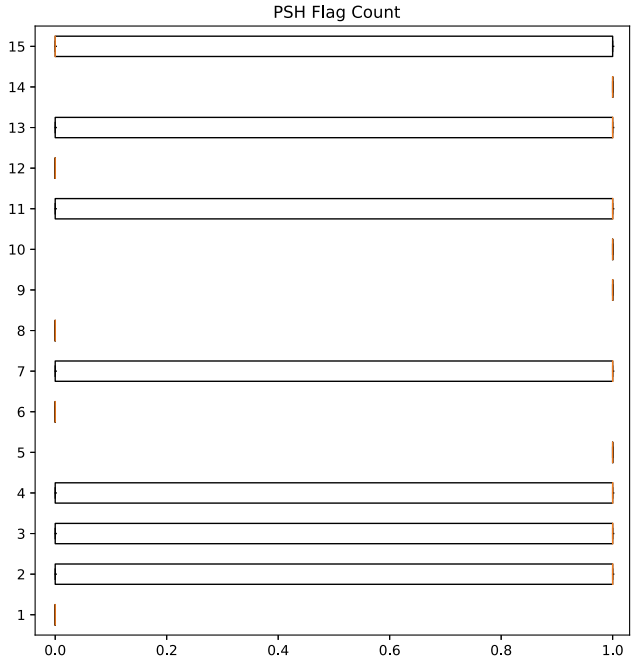


Fig. 6 "PSH Flag Count" range by class

5.5 PSH Flag Count

Figure 6 shows the range of values for "PSH Flag Count" (number of packets with the PSH flag set in the TCP header) across classes. The classes appear to divide into two groups—one where the PSH flag is used and the other where the PSH flag is not. However, for certain attack classes—2, 3, 4, 7, 11, 13, 15—all samples of flows in a class do not appear to use the PSH flag.

5.6 Active Max

Figure 7 shows the range of values for "Active Max" (the maximum time a flow was active before becoming idle) across classes. Here, two attack classes—5 and 12—have significantly different median values compared to the remaining classes. Even so, the range of values for these two classes subsumes those for the other classes.

6 Discussion

Is it possible for machine learning models built over summary data to obtain performance similar to production IDS/IPS? We answer this question by revisiting the top-6 features listed in the section above.

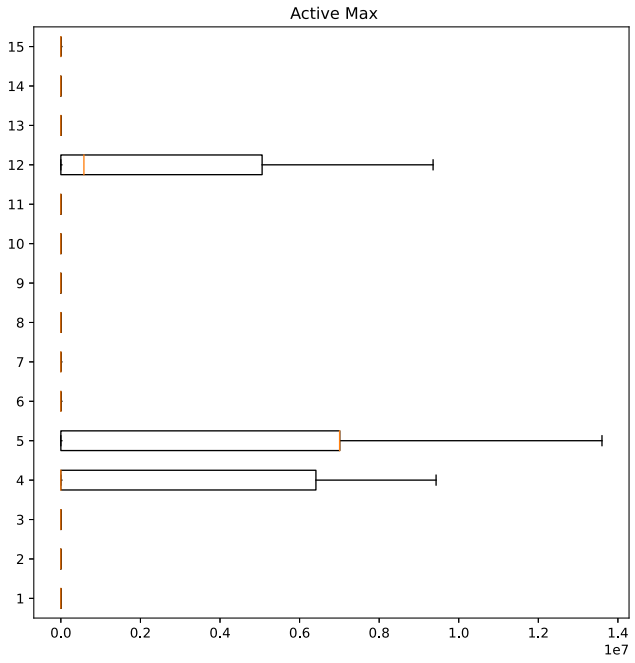


Fig. 7 "Active Max" range by class

6.1 Value of top features

2 of the 6 most important features (PSH Flag Count, Active Max) are the results of software developer choices in normal (non-malicious) applications. The PSH Flag is set when developers want a packet delivered (by the operating system) to the application immediately upon arrival. The Active Max feature captures the longevity of a traffic flow, which in turn depends on the underlying application characteristics that a developer is trying to achieve.

3 of the 6 features (Fwd Packet Length Std, Min Packet Length, Packet Length Variance) are properties of the packet lengths in a traffic flow. Usually, these properties are not chosen directly by an application developer but are the result of other application design choices made by the developer.

Given the nature of these 5 features, we expect them to carry some signal concerning the maliciousness of a traffic flow. However, we do not expect any single one of them to flag a traffic flow as malicious. This intuition is corroborated by the box plots above—none of the 5 features appear to signal malicious traffic flows on its own. Further, all combinations of these features are legitimate application design choices (or results of legitimate choices). It is easy to believe that combinations of these five features would be incorrectly flagged as malicious by machine learning models. That is, machine learning models would overfit the traffic flow samples in the CSV files and yield false positives during production use.

The choice of Destination Port—the remaining top-6 feature—is guided by networking history and norms. Certain ports are reserved for specific applications. Other port ranges are allocated for developer experimentation. Portscans (class 13) scan for a wide range of ports. Such scans have a high probability of being malicious.

However, no single traffic flow (as captured in the CSV files) can list more than one Destination Port per traffic flow—only a single number is allowed in this field. Usually, portscans instantiate multiple traffic flows to multiple Destination Ports to probe for open ports from the same source IP address. Production IDS/IPS and Network Traffic Analysis (NTA) systems can track these flows and flag the scan as malicious [16, 17]. But, the CSV files do not contain adequate information to reliably detect a portscan—they contain no source IP address or timestamps to correlate the flows.

6.2 Realistic performance standard

Given the lack of strong signals in the top 6 features, one would expect mediocre classification performance from the machine learning models trained over the CSV files. Indeed, [1] reports weighted precision performance between 0.77 and 0.98 and recall performance between 0.04 and 0.98.

Even though the precision and recall numbers extracted from [1] are illustrative—they were not presented as adequate for production deployment—examining the highest performance gives us an idea of the "deployability gap" between infosec research models and production systems.

For every 10,000 true positive + false positive flows, a precision of 0.98 would produce 200 false positive flows. Similarly, for every 10,000 true positive + false negative flows, a recall of 0.98 would produce 200 false negative flows. The two types of mislabeling combined will lead network security teams to miss attacks altogether, investigate normal flows as attacks, or mistake one type of attack for another. An IDS system with equivalent performance in a production network will be quickly decommissioned to preserve the security team's effectiveness.

While the normal traffic and attack traffic mix in production networks is likely to be different than that in the CSV files, we can surmise that precision and recall performance would have to be a lot better in large networks with millions of daily traffic flows [18, 19]. In fact, security teams look for near-perfect (0.999, 0.9999, or better) precision and recall for known attacks.

6.3 PCAP-based research

So, what are researchers to do if there is insufficient signal in the CSV files?

There are two fundamental problems with the construction of the CIC-IDS CSV files. First, these CSV files discard some important data (for example, IP addresses) for intrusion detection. Second, the summarization from the underlying traffic flow (PCAP data) to the CSV files is somewhat

arbitrary—the summarization is at the mercy of the features selected in the hand-coded summarization tool. Thus, any machine learning model built on top of the CSV files is handicapped.

Infosec researchers should consider using the PCAP data instead of the CSV files to access the finer-grained information about traffic flows. Perhaps, then, classification performance for known attacks would be good enough to rival that of IDS/IPS currently being used in networks. Researchers must contend with larger file sizes and higher training times in conducting such work. They will likely need to apply some dimensionality reduction techniques, such as principal component analysis (PCA) or t-Distributed Stochastic Neighbor Embedding (t-SNE), to the PCAP data. Algorithmic dimensionality reduction may find a better balance between summarization and feature preservation than is implicitly recorded in the CIC-IDS CSV files.

An alternative way to handle the PCAP data size is to limit the amount of data from each packet (for example, by using only the first 64 bytes). The data truncation would preserve the entirety of the TCP/IP header information but discard some of the (usually less interesting) data payload. Similarly, one can experiment with ignoring all but the first few packets in a traffic flow. These two techniques could result in preserving most of the signal while discarding a significant fraction of the data. As it turns out, commercial IDS/IPS use variations of the above-mentioned data reduction techniques.

Infosec researchers should also consider creating/testing machine learning models that use multiple traffic flows to detect malicious or misbehaving nodes. Most commercial IDS/IPS and NTA systems implement an analytics engine that accommodates multiple flows. Passing a verdict on individual traffic flows without combining available signals across multiple flows is unlikely to push infosec researchers' performance high enough to rival traditional IDS/IPS and NTA systems.

Finally, infosec researchers should attempt to deploy their best-performing machine learning models over a test network to understand their practical usefulness. They could compare the performance of their models with an open-source tool such as Snort [8] or Suricata [9] over the same network.

7 Conclusion

In the sections above, we have discussed machine learning models that use the summary CSV files described in [1] and contrasted the models' performance with production IDS/IPS. We have also discussed a sample of the features available in the CSV files.

We have shown that the features included in the CIC-IDS CSV files are too coarse and incomplete to lead to adequate detection performance in production deployments. As a result, network security teams are unlikely to deploy such intrusion detection systems.

We have also suggested a direction for future infosec research—using PCAP data to train machine learning models and getting a feel for their characteristics in test networks.

The critique of summarized data for intrusion detection in this paper likely applies to other network traffic datasets used in infosec research. A future task entails validating the hypothesis of this paper against datasets beyond CIC-IDS 2017.

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Data Availability The datasets generated during and/or analyzed during the current study are available at https://www.unb.ca/cic/datasets/ids-2017.html.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Appendix A Code

The code to create a new CSV file as described in Sect. 5 is available at https://github.com/r-dube/CICIDS/blob/main/ids_mal.ipynb. The code to run logistic regression and create the figures above is at https://github.com/r-dube/CICIDS/blob/main/ids_mal_viz.ipynb. Both sections of code use Python via Jupyter Notebooks.

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